Course Information				
Course Title	Creative Engineering Design		Course ID	SOC1020
			(Course Type)	Technical core sequence
(Credit/ hours per week)		3/3	version (date)	3.1 (2022.01)

Instructor: Dr. Sarvar Abdullaev

Course Learning Objective

This course is about applying creative engineering decisions to designing products. This course aims at teaching following skills:

	Course Objectives	Relation with Program Outcomes
1	Understanding basic concepts of User Experience (UX) research and design for digital products	4, 9, 11
2	Modelling simple and interactive User Interface (UI) for digital products	3, 4
3	Understanding key steps in product design and development process	3, 4
4	Modelling conceptual design of products and selecting conceptual models for further development	3, 4, 7
5	Prototyping and testing products according to their detailed specifications	1, 6
6	Practicing Lean Startup methodologies in product design	6, 7

Course Description

This course blends the perspectives of marketing, design, and manufacturing into a single approach to product development. As a result, we provide students of all kinds with an appreciation for the realities of industrial practice and for the complex and essential roles played by the various members of product development teams. We attempt to strike a balance between theory and practice through our emphasis on methods and technologies such as Figma and Fusion 360. The methods we present in this course are typically step-by-step procedures for completing tasks. We use different products as the examples for the illustration of different techniques that are commonly used in designing industrial solutions or consumer products. Moreover we are going to teach about user experience (UX) to design digital products. It provides specific tools and tips for learning, designing and shipping amazing products through the application of Lean Startup methodology.

Program Outcomes (PO)

NO	Details
1	An ability to apply knowledge of mathematics, science, and engineering (Basic knowledge)
3	An ability to design a system, component, or process within realistic constraints (a ability to design)
4	An ability to identify, formulate, and solve engineering problems (solve engineering problems)
6	An ability to function on multi-disciplinary teams (Teamwork)
7	An ability to communicate effectively (Communication)
9	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context (understanding of the impact)
11	An understanding of professional and ethical responsibility (Professional Ethics)

(Prerequisites):

(Recommended Courses after This Course): Internet Programming, Application Development in Java

Textbook

Title	Authors	Publisher	Place	Year	ISBN
Product Design and Development (7th ed) (aka PD&D book)	K. Ulrich, S. Eppinger and M. Yang	McGraw Hill	US	2020	978-0073404776

References	Interaction Design: Beyond Human-Computer Interaction (aka IxD book)	J Preece, H Sharp, Y Rogers	John Wiley & Sons, Ltd	US	2019	978-1119547259
	The Design Of Everyday Things: Revised And Expanded Edition	Don Norman	Basic Books	US	2013	978-0465050659
	Laws of UX: Using Psychology to Design Better Products & Services	Jon Yablonski	O'Reilly	US	2020	978-1492055310
	40 Principles Extended Edition: Triz Keys to Technical Innovation	G. Altshuller	Technical Innovation Ctr	US	2005	978-0964074057

Lecture type	Lectures and Classroom activities
Notes	Individual assignments are not graded, but marked for submission. Maximum points for individual assignment is 10%. There are 3 design projects in this course: 2 conceptual design projects 20% each and 1 product design project 40%.

Evaluation Criteria	(Attendance)	10%	(Projects)	80%	(Individual Assignment)	10%		
	(Mid-term Exam)		(Final Exam)		**(Total)**		100 %	

Methods of Evaluation	Assessment will be made on the basis of design projects, individual assignments and attendance.

Weekly Topical Outline of Course		
(1st Week)	Topic	Introduction
	Class details	- What is engineering? - What is interaction design? - What is bad and good design? - Engineering Idea #1: Cellular Automata
	Assignments	• Individual Assignment 1
(2nd Week)	Topic	Fundamentals of Interaction
	Class details	- Affordances, signifiers, mapping, feedback and conceptual model - 7 stages of action - Human errors - FigJam brainstorming
	Assignments	• Read Norman chapters 1-5 • Individual Assignment 2
(3rd Week)	Topic	Customer Needs and User Research
	Class details	- Design thinking process and methodology - Opportunities identification - Identifying customer needs - Qualitative and quantitative user research - Engineering Idea #2: Ant Colony
	Assignments	• Read PD&D chapters 1-5 • Read Norman chapter 6 • Conceptual Design Project 1

(4rd Week)	Topic	Product Specifications
	Class details	- Problem statement - Quality function deployment (QFD) - Establishing target specifications - Design variables - Setting final specifications - Engineering Idea #3: Kidney Exchange
	Assignments	• Read PD&D book chapter 6 • Individual Assignment 3
(5th Week)	Topic	Concept Generation
	Class details	- Analytical vs creative thinking - Impediments to creativity - Strategies for creative thinking - SCAMPER method - Design for validation - Getting started with Figma (Frames, Shapes, Images, Components)
	Assignments	• Read PD&D book chapter 7 • Individual Assignment 4
(6th Week)	Topic	UX/UI Design for Web/Mobile
	Class details	- UI Principles - Laws of UX (Jakob's Law, Fitts's Law, Hick's Law, Miller's Law, Postel's Law) - Diagrams, Sketches and Wireframes in Figma - Prototyping and handoff in Figma
	Assignments	• Read Yablonski's "Laws of UX" book • Individual Assignment 5
(7th Week)	Topic	Concept Selection and Testing
	Class details	- Decision matrix - Concept screening - Concept scoring - Concept testing - Engineering Idea #4: Pricing Options
	Assignments	• Read PD&D book chapters 8 and 9 • Conceptual Design Project 2
(8th Week)	Topic	Mid-term exam.
	Class details	No classes
	Assignments	
(9th Week)	Topic	Prototyping and CAD
	Class details	- Understanding prototyping - Principles of prototyping - Prototyping technologies - Planning for prototypes - Fusion 360 fundamentals and getting started with modeling
	Assignments	• Read PD&D book chapter 14 • Watch Fusion 360 videos (Introduction to Fusion 360 and Get started with Modeling)

(10th Week)	Topic	TRIZ 1
	Class details	- Introduction to TRIZ - Technical contradiction - Contradiction matrix - 40 inventive principles - TRIZ exercises
	Assignments	• Read Altshuller's book • Watch Fusion 360 videos (3D modeling and solid modeling basics) • Individual Assignment 6
(11th Week)	Topic	TRIZ 2
	Class details	- Physical contradictions - Separation principle - TRIZ exercises - Engineering Idea #5: Small World Networks
	Assignments	• Read Altshuller's book • Watch Fusion 360 videos (Mechanical assemblies fundamentals) • Product Design Project
(12th Week)	Topic	Presentations
	Class details	Groups present their projects for Conceptual Design Project 2
	Assignments	
(13th Week)	Topic	Robust Design
	Class details	- What is robust design? - Control factors, noise and performance metrics - Designing experiments - Evaluating experimental results
	Assignments	• Read PD&D book chapter 15
(14th Week)	Topic	Patents, Intellectual Property and Plagiarism
	Class details	- What is intellectual property? - How to protect intellectual property? - Patent types and their application process - What is plagiarism? Implications of plagiarism
	Assignments	• Read PD&D book chapter 16
(15th Week)	Topic	Final exam
	Class details	No classes
	Assignments	
(16th Week)	Topic	Presentations and Demos
	Class details	Groups demonstrate their prototypes for Product Design project.
	Assignments	